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APPENDICES 27

Task A- Analysis of Historical UK Weather Data: Multivariate Regression Analysis

1. Purpose of the Analysis

This analysis examined historical UK weather data to explore relationships between minimum temperature, sunshine hours, and rainfall. It focused on detecting outliers and testing if rainfall can be modelled linearly with temperature, sunshine, and region, aiming to enhance weather forecasting and inform policymakers and environmental research.

2. Methods and Assumptions

Data Examination and Preprocessing:

- The dataset was cleaned on Google Collabs, there were many null values which was cleaned. There were many outliers like the temperature were showing very high values, the sun hours were wrongly imputed. We have cleared all that errors.

Variable	Mean	Std Dev	Minimum	Maximum	N
rain	67.4928372	38.2156345	0	179.0000000	22254
tmin	6.6571807	3.7082964	-1.6000000	16.8000000	22254
sun	124.8721398	62.3822060	2.8000000	314.4000000	22254

Fig.1

- The dataset contained 22,254 observations (Fig.1) with variables including year, month, maximum and minimum temperature (tmax, tmin), air frost days (af), rainfall (rain), sunshine hours (sun), and station (region).
- Boxplots (Fig.2) were utilized to detect outliers in key continuous variables (rain, tmin, sun). Significant outliers were present in rainfall, a typical characteristic in environmental data.

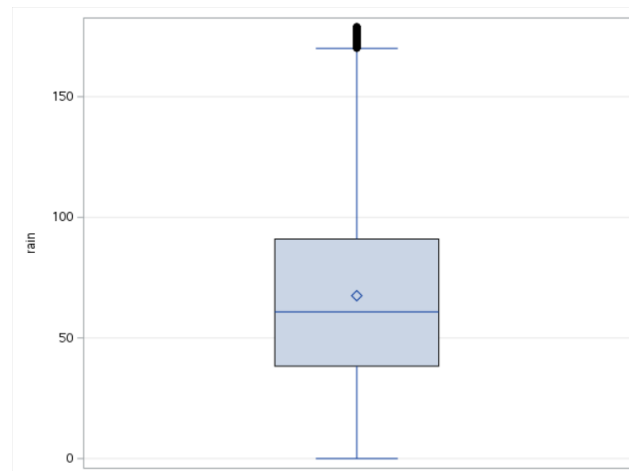


Fig.2

Correlation

The Pearson Correlation Coefficients show weak negative relationships between rain and the variables: tmax (-0.19814), tmin (-0.06645), and sun (-0.37778). The strongest inverse correlation is with sunshine hours, suggesting more sunshine is moderately associated with less rainfall. Other relationships are very weak(Fig.3)

3 With Variables:	tmax tmin sun
1 Variables:	rain

Pearson Correlation Coefficients, N = 22254	
	rain
tmax	-0.19814
tmin	-0.06645
sun	-0.37778

Fig.3

Multivariate Linear Regression Model

- The response variable was rainfall, and predictor variables included minimum temperature (tmin), sunshine hours (sun), and station (treated as a categorical variable).
- The model was built using the Ordinary Least Squares (OLS) method.
- A square root transformation was applied to the rainfall variable to stabilize variance and improve model performance.

Model Assumptions Checked

- **Linearity:** The residual plot (Fig.4) shows a random scatter around zero with no distinct pattern, indicating the linearity assumption is satisfied. The spread of residuals is consistent, suggesting homoscedasticity. Although a few outliers are present, they appear minimal. Overall, the linear model is appropriate for predicting rain.

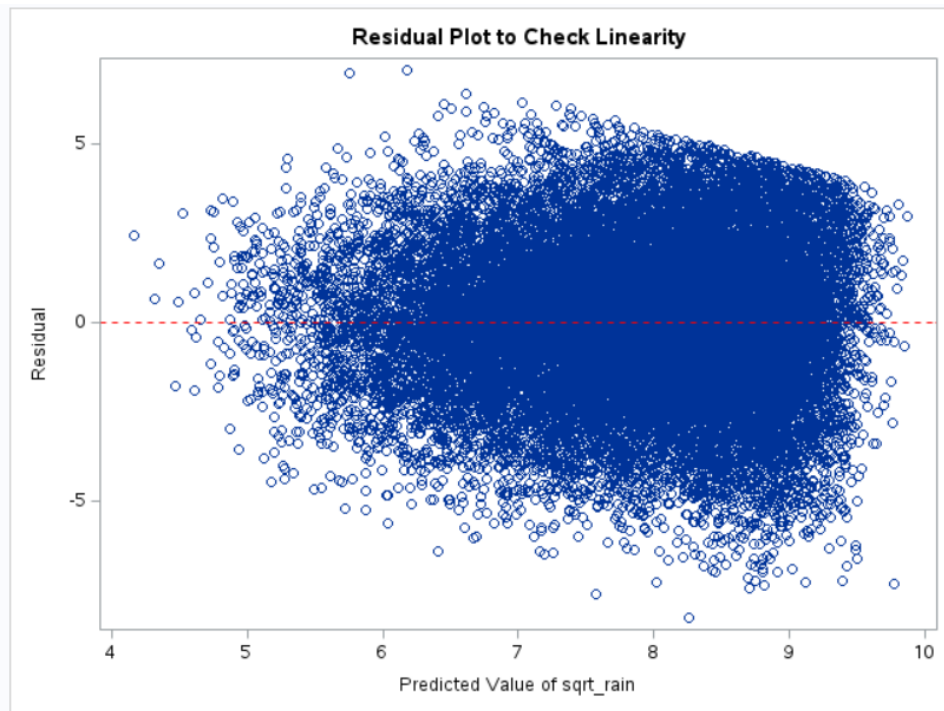


Fig.4

- **Homoscedasticity:** Homoscedasticity was not seen while examining the residual plot(Fig.5).

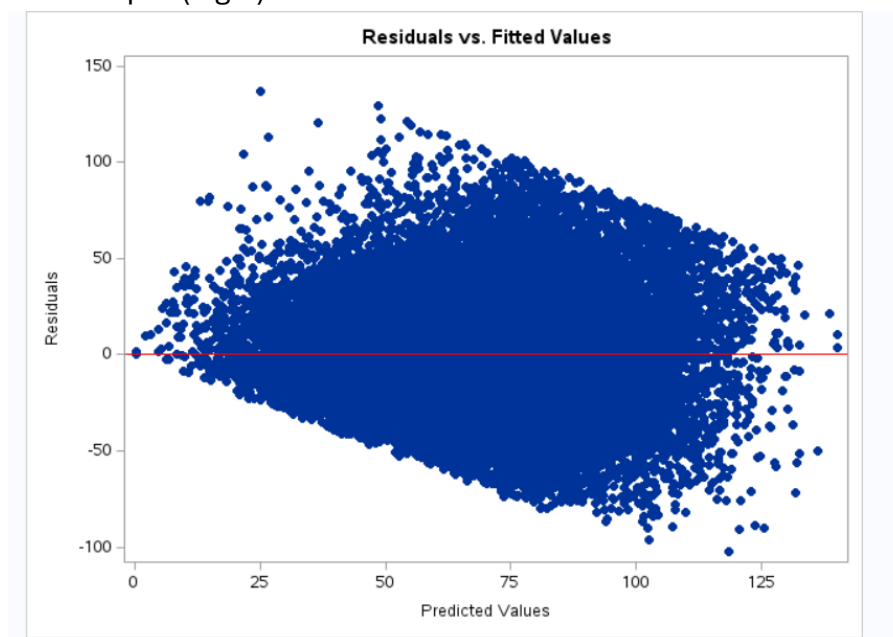


Fig.5

We did a square root transformation on the rain variable in SAS. No heteroscedasticity was observed after applying the square root transformation, confirming constant variance of residuals(Fig.6)

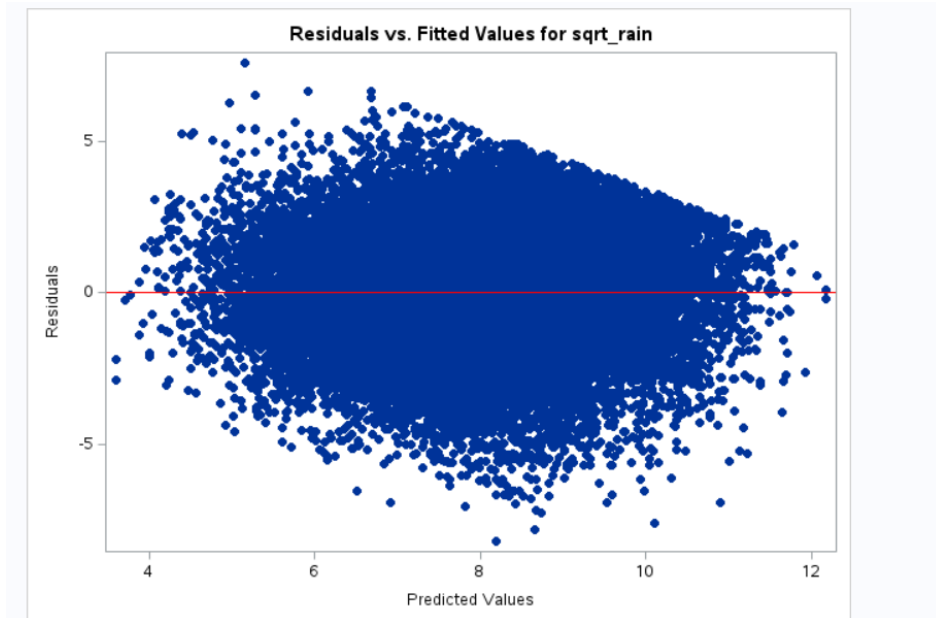


Fig.6

- **Multicollinearity:** Variance Inflation Factor (VIF) was used to detect multicollinearity among predictors.

Coeff Var		27.38512		
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Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Tolerance	Variance Inflation
Intercept	1	9.32256	0.03400	274.16	<.0001	.	0
tmin	1	0.16672	0.00487	34.22	<.0001	0.63769	1.56815
sun	1	-0.02060	0.00028966	-71.11	<.0001	0.63769	1.56815

Fig.7

There is no multicollinearity in this model (Fig.7) because VIFs are well below 5. Tolerance values are comfortably above 0.1 which suggests no multicollinearity. VIF values < 5 indicate no multicollinearity. Since both VIFs are 1.56815, there is no concern for multicollinearity between tmin and sun. Both tmin and sun are statistically significant predictors. Both tmin and sun are highly significant predictors ($p < 0.0001$) based on their t-values.

Normality of Residuals: The Q-Q plot (Fig.8) shows that the residuals mostly align with the normal line, indicating they are approximately normally distributed. Slight deviations at the tails suggest minor departures from normality, but overall, the model's assumption of normally distributed residuals appears valid. Further investigation may be optional.

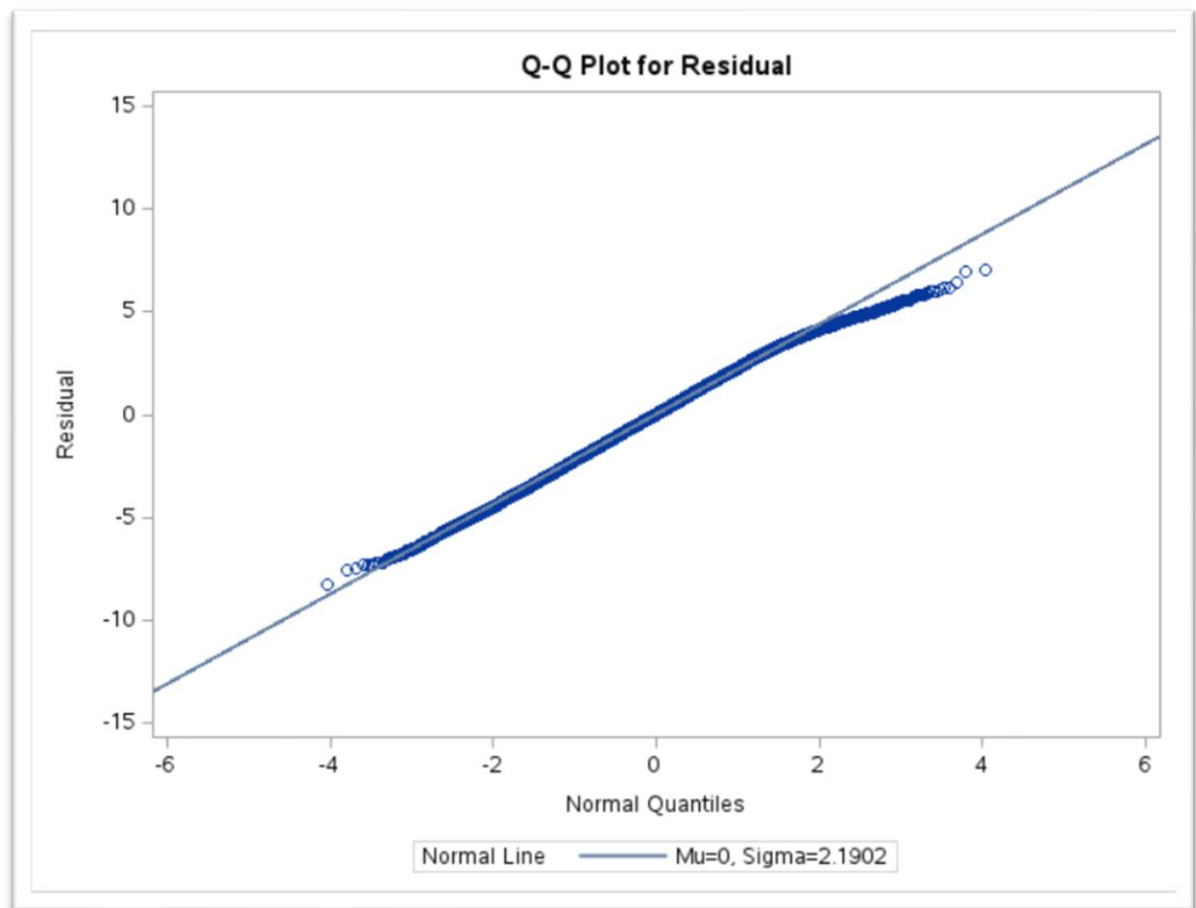


Fig.8

Independence of Errors: The Durbin-Watson statistic(Fig.9) was **1.624**, indicating that residuals are approximately independent with minimal autocorrelation. The 1st order autocorrelation was **0.188**, further supporting this conclusion.

The REG Procedure
Model: MODEL1
Dependent Variable: sqrt_rain

Durbin-Watson D	1.624
Pr < DW	<.0001
Pr > DW	1.0000
Number of Observations	22254
1st Order Autocorrelation	0.188

Fig.9

This value is slightly below 2, suggesting a mild positive autocorrelation in the residuals. However, the value is close enough to 2 to indicate that autocorrelation is not a serious issue.

The 1st order autocorrelation of 0.188 further suggests a low level of correlation between consecutive residuals. The mild autocorrelation is unlikely to significantly distort the regression results, meaning the model's predictions remain reliable.

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > t
Intercept	1	8.916246	0.104765	85.11	<.0001
tmin	1	0.163485	0.004590	35.61	<.0001
sun	1	-0.019692	0.000273	-72.00	<.0001
station aberporth	1	0.649754	0.124734	5.21	<.0001
station armagh	1	0.199338	0.118688	1.68	0.0931
station bradford	1	0.260175	0.118242	2.20	0.0278
station braemar	1	0.744113	0.151818	4.90	<.0001
station camborne	1	1.117899	0.136730	8.18	<.0001
station cambridge	1	-0.923847	0.133325	-6.93	<.0001
station cardiff	1	1.187316	0.180854	6.57	<.0001
station cwmystwyt	1	2.692936	0.151120	17.82	<.0001
station dunstaffn	1	1.972962	0.198564	9.94	<.0001
station durham	1	-0.546769	0.118686	-4.61	<.0001
station eastbourn	1	0.394094	0.127109	3.10	0.0019
station eskdalemu	1	2.536950	0.137023	18.51	<.0001
station heathrow	1	-0.730663	0.124530	-5.87	<.0001
station hurn	1	0.699850	0.132748	5.27	<.0001
station lerwick	1	1.259557	0.119831	10.51	<.0001
station leuchars	1	-0.045970	0.127087	-0.36	0.7176
station lowestoft	1	-0.629783	0.122142	-5.16	<.0001
station manston	1	-0.560922	0.132267	-4.24	<.0001
station nairn	1	-0.564139	0.121801	-4.63	<.0001
station newtonrig	1	0.849460	0.178483	4.76	<.0001
station oxford	1	-0.340080	0.118809	-2.86	0.0042
station paisley	1	1.290380	0.132481	9.74	<.0001
station ringway	1	0.318985	0.134846	2.37	0.0180
station rossonweye	1	-0.086559	0.121397	-0.71	0.4758
station shawbury	1	-0.247879	0.126420	-1.96	0.0499
station sheffield	1	0.037685	0.118248	0.32	0.7500
station southampt	1	0.308066	0.119499	2.58	0.0099
station stornoway	1	1.399681	0.118786	11.78	<.0001
station tiree	1	1.629043	0.118713	13.72	<.0001
station valley	1	0.541661	0.117667	4.60	<.0001
station waddingto	1	-0.541035	0.123150	-4.39	<.0001

Fig.10

Least Squares Model (No Selection)					
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	35	38882	1110.91547	279.90	<.0001
Error	22218	88184	3.96904		
Corrected Total	22253	127066			

Root MSE	1.99224
Dependent Mean	7.86022
R-Square	0.3060
Adj R-Sq	0.3049
AIC	52970
AICC	52970
SBC	31002

Fig.11

3.Model Performance

- The R-squared value(Fig.12) of 0.311 indicates that approximately 31.1% of the variance in rainfall is explained by the model.
- Adjusted R-squared of 0.310 suggests a moderately fitting model.
- F-statistic of 286.9 with a p-value < 0.0001 confirms the model is statistically significant.

Impact of Predictors:

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > t
Intercept	1	9.322565	0.034004	274.16	<.0001
tmin	1	0.166724	0.004873	34.22	<.0001
sun	1	-0.020599	0.000290	-71.11	<.0001

Fig.12

- **Minimum Temperature (tmin):** Coefficient = 2.5428 (p < 0.0001). A positive relationship was observed, indicating that higher temperatures are associated with increased rainfall. For every 1-unit increase in tmin, the dependent variable increases by **0.16672** units, holding other variables constant.

- **Sunshine Hours (sun):** Coefficient = -0.3106 ($p < 0.0001$). This negative association suggests that more sunshine leads to reduced rainfall. For every 1-unit increase in sun, the dependent variable decreases by **0.02060** units, holding other variables constant. Tmin has a positive impact, while sun has a negative impact.
 - **Station (Region):** Significant differences in rainfall across regions were detected. For instance, stations like cwmystwyth and eskdalemuir reported higher rainfall, whereas cambridge and heathrow had lower levels.
-

4. Conclusion

The multivariate regression model validated the hypothesis that rainfall can be explained by minimum temperature, sunshine hours, and region. Key conclusions include:

- **Model Effectiveness:** The model is statistically significant and moderately effective, explaining 31.1% of the variation in rainfall. However, 68.9% of the variance remains unexplained, suggesting additional factors influence rainfall.
- **Predictor Influence:** Higher temperatures correlate positively with rainfall, whereas increased sunshine hours correlate negatively.
- **Regional Variation:** There is substantial variability in rainfall across different UK regions.
- **Model Improvements:** The square root transformation of the rainfall variable successfully addressed issues with heteroscedasticity, improving model reliability.

For every 1°C increase in minimum temperature, the square root of rainfall increases by approximately **2.54 units**, indicating that warmer temperatures are associated with more rainfall. Conversely, for every additional hour of sunshine, the square root of rainfall decreases by approximately **0.31 units**, suggesting that increased sunshine is linked to lower rainfall.

Recommendations:

- **Include Additional Variables:** Incorporating other meteorological factors like wind speed, humidity, and atmospheric pressure might enhance model performance.
- **Explore Non-Linear Models:** Advanced models such as Generalized Linear Models (GLM) or machine learning techniques could better capture complex relationships.

In summary, this analysis provides valuable insights into how temperature, sunshine, and geographic region influence rainfall patterns in the UK. While the model demonstrates statistical significance, further refinement and additional predictors are necessary to improve predictive accuracy.

Task B- Market Factor Analysis Report (PCA)

1.Purpose of the Analysis

The objective of this analysis is to identify underlying factors influencing consumer behaviour regarding product purchases and household characteristics. Principal Component Analysis (PCA) was used as the extraction method, followed by Varimax rotation to achieve a clearer factor structure. The data was cleaned using Google Collab.

2.Eigenvalues and Scree Plot Analysis

From the Eigenvalues table (Fig.13), the first factor has an eigenvalue of **5.0656**, explaining **46.05%** of the variance. The second factor has an eigenvalue of **1.4126**, explaining an additional **12.84%**. According to **Kaiser's criterion**, which suggests retaining factors with eigenvalues greater than 1, two factors were retained. The scree plot (Fig.14) shows a sharp drop after the second factor, supporting this decision.

Initial Factor Method: Principal Components

Prior Communality Estimates: ONE

Eigenvalues of the Correlation Matrix: Total = 11 Average = 1				
	Eigenvalue	Difference	Proportion	Cumulative
1	5.06558512	3.65295486	0.4605	0.4605
2	1.41263026	0.61884384	0.1284	0.5889
3	0.79378642	0.10205837	0.0722	0.6611
4	0.69172805	0.08218253	0.0629	0.7240
5	0.60954552	0.05789689	0.0554	0.7794
6	0.55164863	0.11351899	0.0501	0.8295
7	0.43812964	0.03223254	0.0398	0.8694
8	0.40589710	0.01265280	0.0369	0.9063
9	0.39324430	0.04010104	0.0357	0.9420
10	0.35314326	0.06848157	0.0321	0.9741
11	0.28466169		0.0259	1.0000

11 factors will be retained by the NFACTOR criterion.

Fig.13

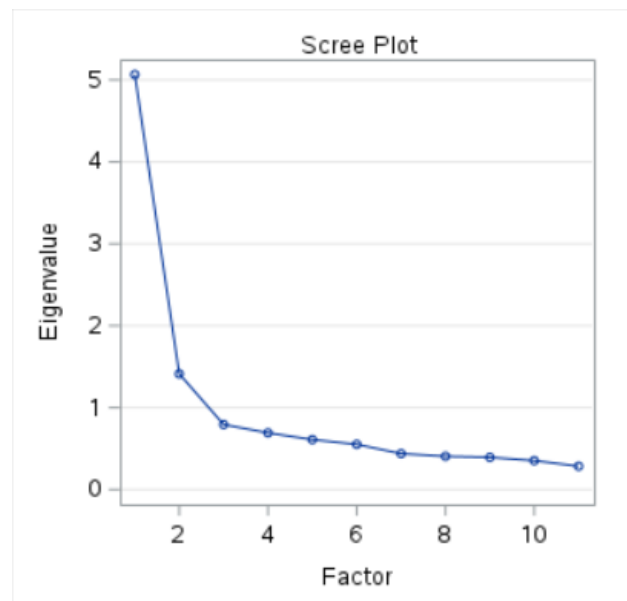


Fig.14

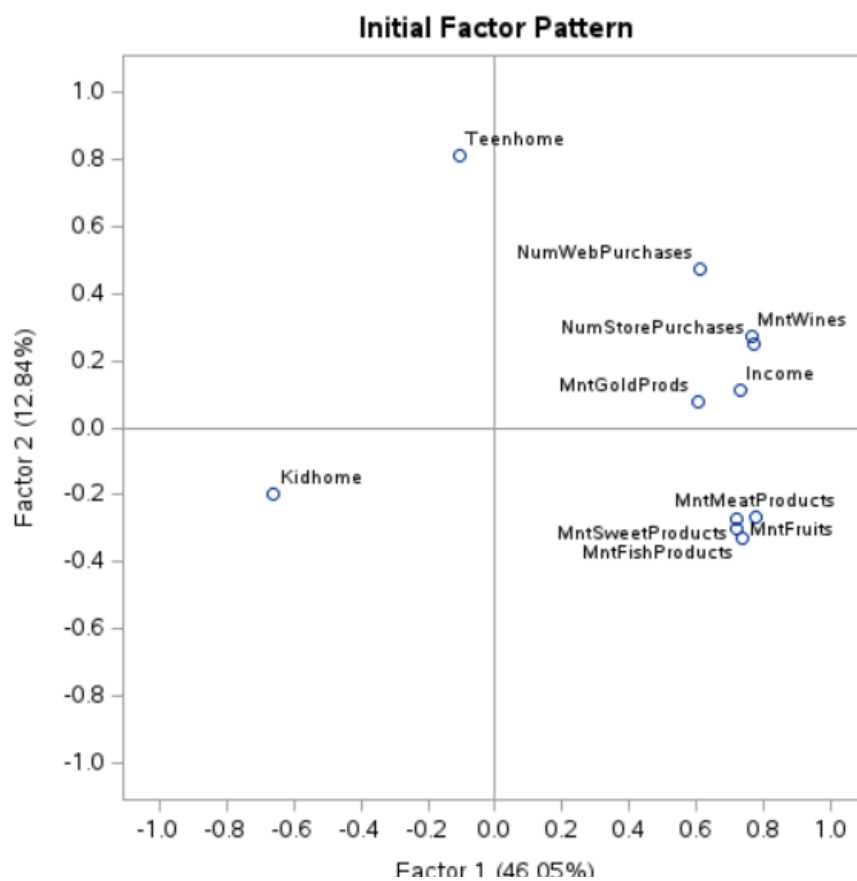


Fig.15

Factor Pattern			
		Factor1	Factor2
Income	Income	0.72954	0.11169
Kidhome	Kidhome	-0.65972	-0.19767
Teenhome	Teenhome	-0.10539	0.80786
MntWines	MntWines	0.76483	0.26995
MntFruits	MntFruits	0.71884	-0.30294
MntMeatProducts	MntMeatProducts	0.77623	-0.26785
MntFishProducts	MntFishProducts	0.73441	-0.32824
MntSweetProducts	MntSweetProducts	0.71624	-0.27087
MntGoldProds	MntGoldProds	0.60423	0.07792
NumWebPurchases	NumWebPurchases	0.61294	0.47221
NumStorePurchases	NumStorePurchases	0.76787	0.24877

Fig.16

By examining the factor loading table (Fig.16), it can be seen that the majority of the variables are loaded on the first factor, this is not what we require, hence indicates that a rotation is required. A **Varimax rotation** is recommended to clarify the factor structure. This will help in better distinguishing between the factors and improving interpretability.

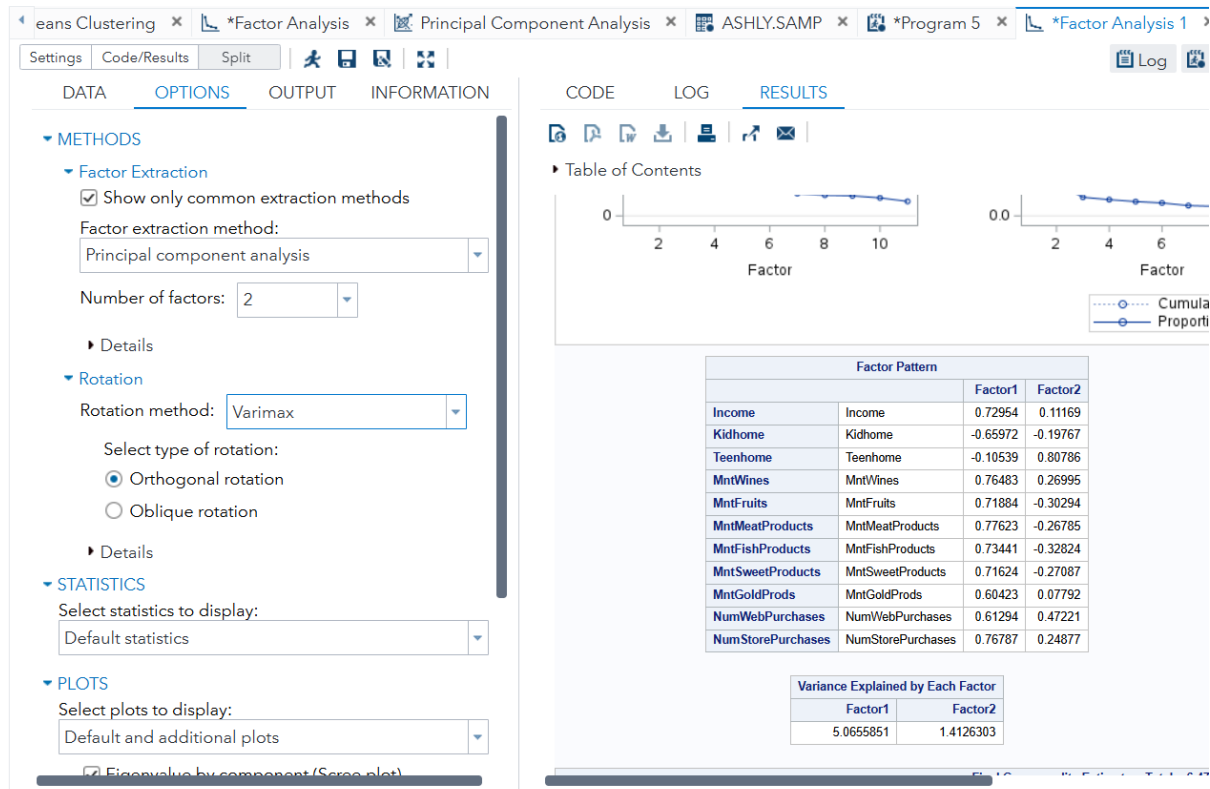


Fig.17

We can see that after applying Varimax rotation, the factor structure (Fig.18,19) became clearer, reducing cross-loadings and enhancing interpretability. Factor 1 now distinctly captures spending behaviour with strong loadings on income and product purchases, while Factor 2 reflects household composition and specific product preferences, notably influenced by the presence of teenagers and food-related expenditures.

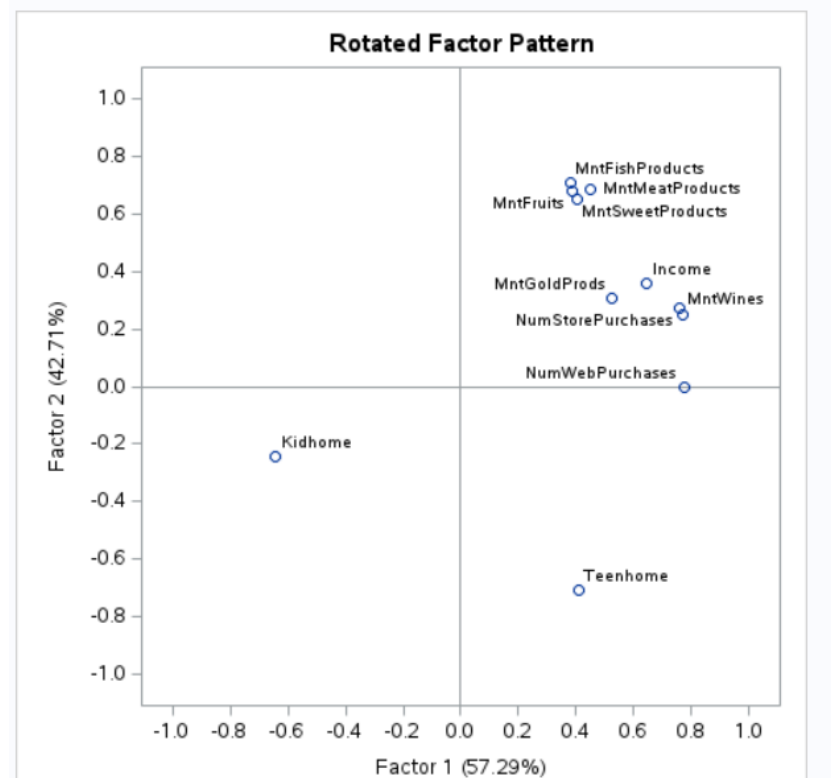


Fig.18

Rotated Factor Pattern			
		Factor1	Factor2
Income	Income	0.64672	0.35562
Kidhome	Kidhome	-0.64368	-0.24490
Teenhome	Teenhome	0.40831	-0.70501
MntWines	MntWines	0.77107	0.25156
MntFruits	MntFruits	0.38576	0.67801
MntMeatProducts	MntMeatProducts	0.45265	0.68511
MntFishProducts	MntFishProducts	0.38271	0.70756
MntSweetProducts	MntSweetProducts	0.40323	0.65099
MntGoldProds	MntGoldProds	0.52675	0.30611
NumWebPurchases	NumWebPurchases	0.77374	-0.00136
NumStorePurchases	NumStorePurchases	0.76058	0.27022

Fig.19

On checking the variance (Fig.20) explained by each factor we can see that the 2 factors account for 6.5 out of 11 variables, which is 59%.

Variance Explained by Each Factor	
Factor1	Factor2
3.7112209	2.7669945

Fig.20

Factor 1 reflects consumers with higher purchasing power and an inclination to spend on a variety of products, particularly luxury and consumable items.

Factor 2 captures how family structure influences spending, with smaller or childless households favouring specific food products.

Factor1 – Spending Behaviour

Factor 2 – Household Preferences

We now need to analysis these factors by the variables, ‘kidhome’ and ‘teenhome’. This is to see whether households having different number of kids and teens put more emphasis on spending on needs or on products.

3.Factor Scores

The boxplot1(Fig.21) shows that households with no children (kidhome = 0) have higher and more varied Factor 2 scores, indicating stronger preferences for specific products. In contrast, households with one or two children (kidhome = 1 or 2) have lower and more consistent scores for both Factor 1 and Factor 2, suggesting more stable and cautious spending behaviour across product categories.

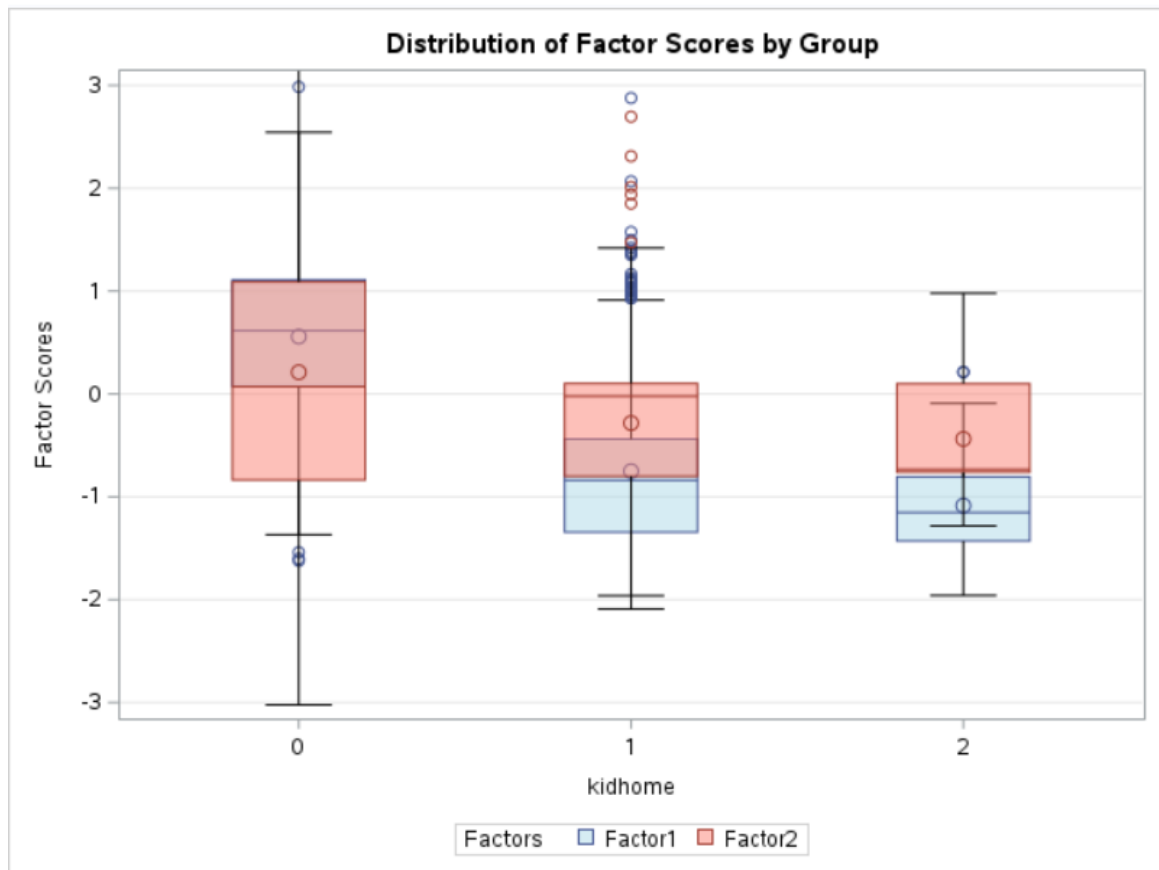


Fig.21

The boxplot2(Fg.22) reveals that households with two teenagers (teenhome = 2) exhibit high **Factor 1** scores, indicating increased spending and purchasing behavior, while their **Factor 2** scores are notably low, suggesting less interest in specific product preferences. In contrast, households with no (0) or one teenager (1) display more balanced and moderate scores across both factors, reflecting more stable consumption habits.

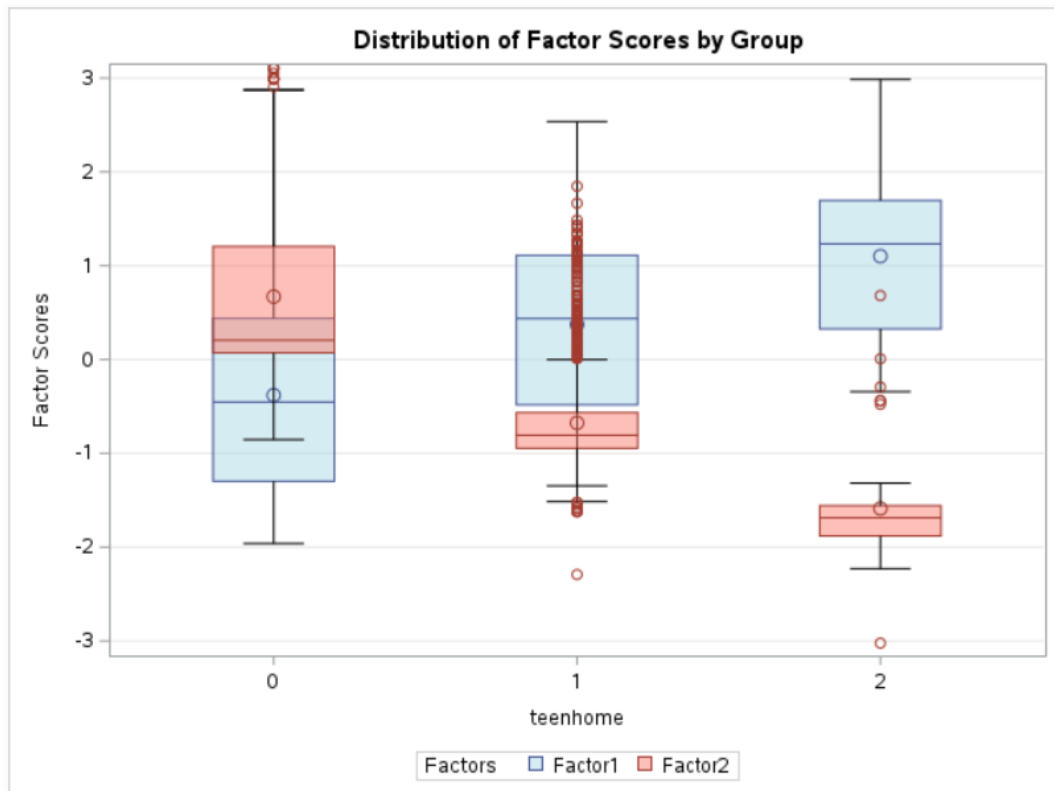


Fig.22

4. Conclusion

The factor analysis reveals two distinct underlying factors influencing the dataset. Factor 1 shows varied distributions across groups, suggesting diverse behavioural or demographic patterns, while Factor 2 displays moderate variance. These factors collectively highlight significant patterns and relationships, aiding in better understanding of group-specific characteristics and trends.

Task C- Employee Satisfaction and Motivation Analysis Report (Cluster Analysis)

1. Purpose of the Analysis

The primary objective of this analysis is to uncover distinct groups of employees within the company based on their satisfaction levels across multiple factors, including Job Satisfaction, Pay Satisfaction, Work-Life Balance, and Team Dynamics. By identifying these groups through cluster analysis, the company aims to better understand employee motivation and satisfaction patterns, enabling the development of targeted strategies to improve employee engagement and retention.

2. Methods Used for Data Analysis

Clustering Approach

The analysis utilized K-Means clustering to segment employees into distinct groups based on satisfaction scores. Five key variables were used for clustering:

- **Satisfaction Level**
- **Number of Projects**
- **Average Monthly Hours**
- **Time Spent at the Company**
- **Salary**

Cluster analysis was conducted for different numbers of clusters (ranging from 2 to 24) to determine the optimal number of groups. The Elbow Method and evaluation of cluster performance metrics such as the Root Mean Square (RMS) Standard Deviation, Maximum Distance from Seed to Observation, and Distance Between Cluster Centroids were used to guide this decision.

3. Feature Analysis

Post-clustering, Cluster Means and Cluster Standard Deviations were examined to understand the central tendencies and variabilities of satisfaction-related variables within each cluster. Additionally, statistical tests (e.g., ANOVA) were implied to assess significant differences between clusters across satisfaction factors.

21	CL54	CL74	903	0.0054	.757	2327	564	
20	CL46	CL45	861	0.0056	.751	2376	278	
19	CL26	CL53	1517	0.0078	.743	2407	559	
18	CL60	CL27	854	0.0083	.735	2442	348	
17	CL106	CL29	1131	0.0085	.726	2485	695	
16	CL28	CL25	1993	0.0088	.718	2537	637	
15	CL36	CL34	630	0.0091	.708	2600	292	
14	CL21	CL23	2249	0.0100	.698	2669	613	
13	CL24	CL32	2248	0.0108	.688	2749	773	
12	CL15	CL30	1097	0.0141	.674	2811	332	
11	CL18	CL38	1531	0.0172	.656	2862	728	
10	CL129	CL33	2064	0.0209	.635	2903	3598	
9	CL13	CL20	3109	0.0234	.612	2956	1118	
8	CL11	CL22	1839	0.0252	.587	3042	645	
7	CL19	CL9	4626	0.0263	.561	3187	1035	
6	CL16	CL14	4242	0.0305	.530	3382	1555	
5	CL17	CL12	2228	0.0357	.494	3665	959	
4	CL10	CL6	6306	0.0917	.403	3369	3952	
3	CL8	CL5	4067	0.1033	.299	3203	1948	
2	CL4	CL7	10932	0.1104	.189	3494	3160	
1	CL2	CL3	14999	0.1889	.000	.	3494	

Fig.25

Cluster 1 shows moderate satisfaction (0.60) with balanced workload and salary (0.45). Cluster 2 has the lowest satisfaction (0.25) but works the most hours with a low salary (0.13). Cluster 3 reflects moderate satisfaction (0.44) with the highest salary (0.73) and longest tenure. Cluster 4 has the highest satisfaction (0.78) but the lowest salary (0.04).

Cluster Means					
Cluster	satisfaction_level	number_project	average_monthly_hours	time_spend_company	salary
1	0.6044250568	0.2391602144	0.3984869443	0.1143723943	0.4547349613
2	0.2554521809	0.5862745098	0.6663197989	0.1960346639	0.1293767507
3	0.4447509937	0.3930212766	0.5411692185	0.4810638298	0.7323404255
4	0.7775371383	0.3919096896	0.5051102944	0.2153692380	0.0413922860

Fig.26

The analysis of statistics of variables (Fig.27) reveals that salary ($R^2 = 0.491251$) and satisfaction level ($R^2 = 0.439264$) are the most influential variables in differentiating clusters, indicating strong explanatory power. Conversely, average monthly hours ($R^2 = 0.182486$) contribute the least. The overall R^2 of 0.365368 suggests moderate variance explanation, guiding focus on salary and satisfaction for impactful decision-making.

Statistics for Variables				
Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
satisfaction_level	0.27322	0.20461	0.439264	0.783370
number_project	0.24652	0.21004	0.274181	0.377753
average_monthly_hours	0.23338	0.21103	0.182486	0.223221
time_spend_company	0.18252	0.15472	0.281585	0.391954
salary	0.31859	0.22726	0.491251	0.965607
OVER-ALL	0.25483	0.20303	0.365368	0.575717

Fig.27

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	6716	0.2058	0.9379		4	0.4957
2	2856	0.2140	0.8354		4	0.5869
3	1175	0.2505	0.9713		1	0.5301
4	4252	0.1743	0.9037		1	0.4957

Fig.28

To visualize this graphically, return to SAS and create a new dataset for these clusters. Navigate to the Output tab and select the first option to save the output to your work file. After selecting this, re-run the analysis. The results will remain the same, but a new dataset will be generated for visualization purposes.

```
CODE LOG RESULTS
1 proc sgplot data=ashly.clusterscore1;
2 scatter x=salary y=number_project/ group=Cluster markerattrs=(symbol=circlefilled)
3 transparency=0.2;xaxis label="salary";
4 yaxis label="number_project";
5 keylegend / title="Cluster";
6 run;
```

Fig.29

Scatter Plots

The scatter plot (Fig.30) shows distinct clustering patterns between satisfaction_level and average_monthly_hours. Cluster 4 (red) mainly covers high satisfaction and diverse working hours, while Cluster 1 (blue) is concentrated in low satisfaction with moderate hours. Clusters 2 and 3 (green and brown) are scattered across lower satisfaction levels, highlighting how satisfaction level significantly influences cluster formation.

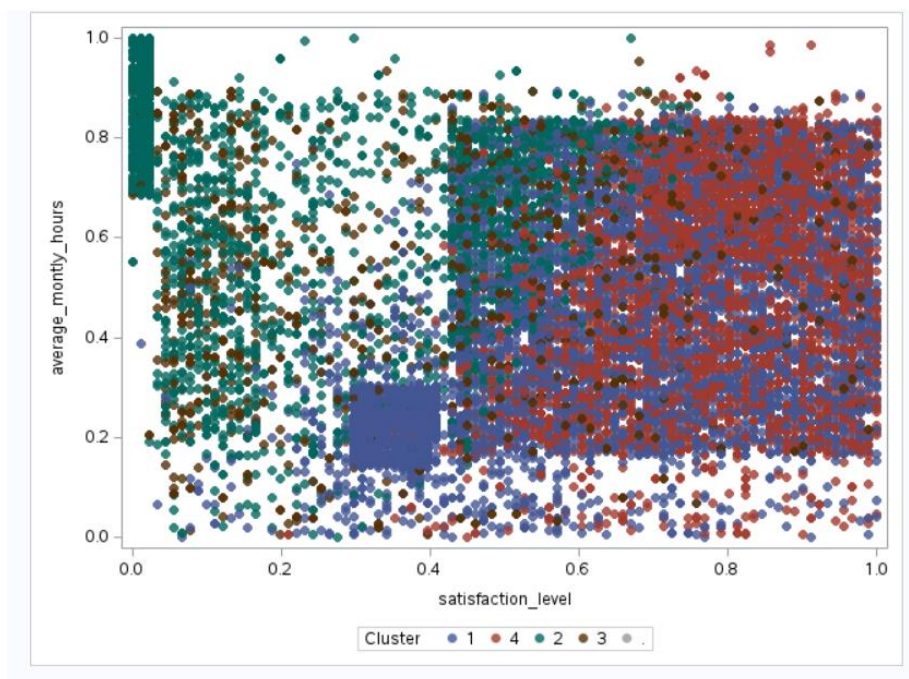


Fig.30

The scatter plot (Fig.31) plot illustrates the relationship between satisfaction_level and salary across clusters. Salaries are distinctly grouped into three levels (0 = low, 0.5 = medium, 1 = high). All clusters are evenly distributed across satisfaction levels, indicating that salary alone does not significantly impact satisfaction. Clustering appears uniform, suggesting other variables may drive cluster differentiation.

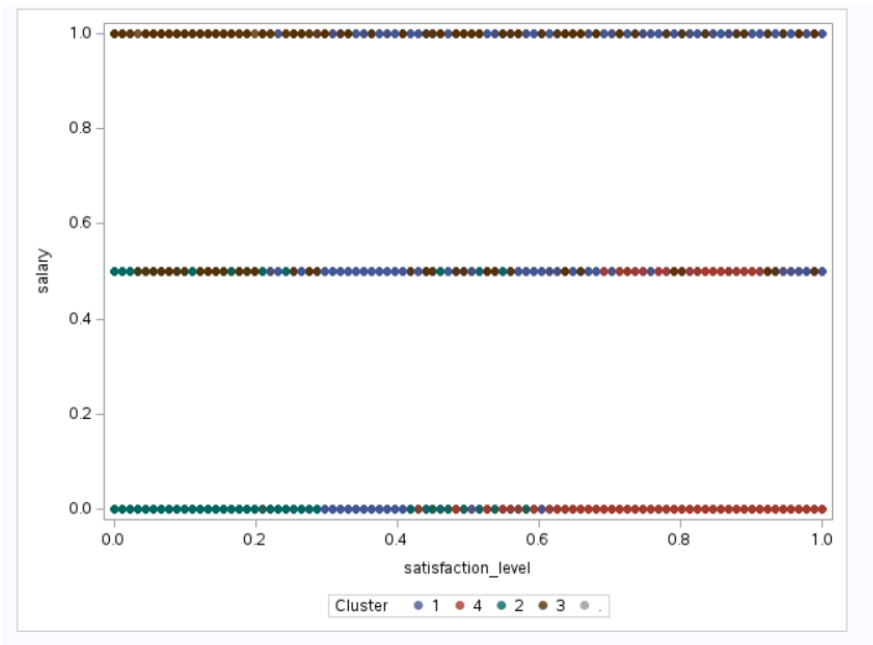


Fig.31

The scatter plot (Fig.32) illustrates the relationship between salary and number_project across clusters. Data points are concentrated at distinct salary levels (low, medium, high) with no clear trend across project counts. Cluster distribution appears uniform, suggesting that salary does not significantly influence the number of projects assigned. Other variables likely contribute more to cluster differentiation.

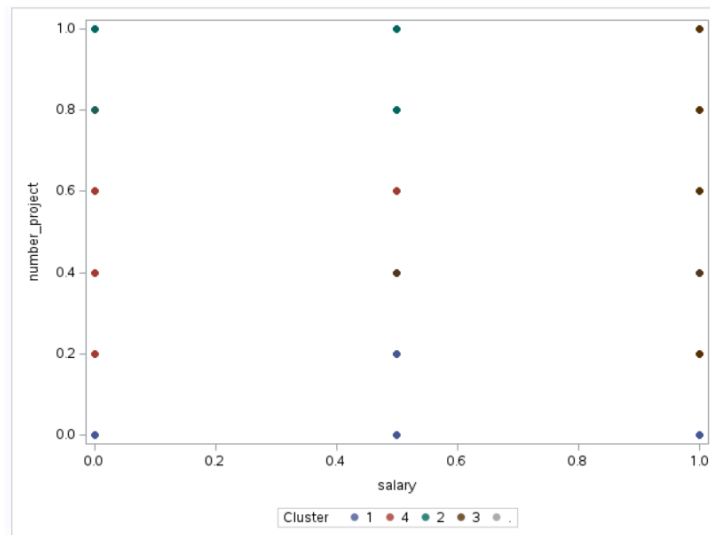


Fig.32

4. Conclusion

The analysis shows that employees' salaries are divided into clear levels—low, medium, and high but salary does not seem to affect how many projects employees work on. Workers at all salary levels handle similar workloads. This suggests that how much someone is paid may not reflect how much work they do. Other factors, like how satisfied employees are or how long they've been with the company, might better explain differences among employee groups. This insight could help improve how tasks and salaries are managed to keep employees motivated and compensated.

APPENDICES

1. This code was used to make residual plot(Fig.4)

```
PROC REG DATA=ashly.square;  
MODEL sqrt_rain = tmax sun;  
OUTPUT OUT=residuals_data P=Predicted R=Residual;  
RUN;  
QUIT;  
  
/* Plot Residuals vs. Predicted Values */  
PROC SGPLOT DATA=residuals_data;  
SCATTER X=Predicted Y=Residual;  
REFLINE 0 / AXIS=Y LINEATTRS=(COLOR=red PATTERN=shortdash);  
TITLE 'Residual Plot to Check Linearity';  
RUN;
```

2. This code was used to make scatter plots during cluster analysis. (Fig.30,31,32)

```
proc sgplot data=ashly.cluster;  
scatter x=salary y=satisfaction_level / group=Cluster markerattrs=(symbol=circlefilled)  
transparency=0.2;  
xaxis label="salary";  
yaxis label="satisfaction_level";  
keylegend / title="Cluster";  
run;
```

3. Google collabs – Most of the data cleaning was done using Google Collabs.